

Derbyshire Shared Care Pathology Guidelines

Glucose Tolerance Test for Primary Care

HbA1c is the preferred first line test for diabetes since approval by WHO for diagnosis, rather than just monitoring of diabetes in 2011. In some individuals, diabetes may also be identified by the presence of raised venous plasma glucose (either on more than one occasion or accompanied by relevant symptoms).

However, HbA1c is contraindicated for diagnosis of diabetes in some situations – please see the Derbyshire Shared Care Pathology guideline ‘Diagnosis of type 2 Diabetes Mellitus using HbA1c’ at <https://www.uhdb.nhs.uk/shared-care-pathology-guidelines>

There may also be other situations where diabetes may be developing rapidly where OGTT is the preferred test to assess for the presence of diabetes OR where staging of diabetes is required.

All glucose values in this document refer to venous plasma glucose levels. Glucose results obtained from “near patient” testing strips should not be used in the diagnosis of DM.

INDICATIONS

Diagnosis of diabetes mellitus:

1. Individuals where a HbA1c is known or likely to be abnormal; most commonly due to problems with red-cell turnover e.g. sickle-cell, thalassemia, liver disease
2. Individuals with antibody positivity and in stage 1 or 2 pre-symptomatic type 1 diabetes in whom the test may be used for staging the disease and guiding appropriate treatments
3. In cystic fibrosis

CONTRAINDICATIONS

None

SIDE EFFECTS

Occasionally nausea, vomiting or diarrhoea as the glucose drink is hyperosmolar.

PREPARATION

An OGTT is an outpatient procedure carried out in the Phlebotomy Department at the FNCH or RDH. Patients must make have appointment before 10am (made using Swiftqueue). Some GP surgeries can also carry out the procedure

Patient

- The patient should maintain a normal diet for three days prior to the test.
- All medication the patient is receiving should be noted on the request form. Drugs such as oral contraceptives, steroids, thiazide and loop diuretics may impair glucose tolerance.

- **The patient should fast from 10p.m. the previous evening**, to give a minimum 10 hour fasting period, and not eat or drink anything other than water, until the test is completed.

Equipment

a) Polycal

GP surgeries can obtain Polycal from pathology by emailing dhft.pathsupplies@nhs.net
Polycal is a carbohydrate drink based on maltodextrin, a partial hydrolysate of corn starch. It is supplied by Cow and Gate in 200 mL bottles. Only 113 ml is required for each patient. This is equivalent to 75g anhydrous glucose. If 3 patients are being tested 2 bottles are sufficient. Measure 113 mL Polycal into a special beaker, add water up to 200 mL mark. Secure plastic cap firmly onto beaker, shake to mix. Polycal is now ready.

Note: A further 100mL of water must be drunk by the patient to make the final volume 300mL

b) For use if the patient has an allergy to citrus flavouring

In exceptional circumstances, when a patient has an allergy to the lemon flavouring, a 'Polycal neutral liquid' is available but prior notice may be required to obtain this.

c) Specimen tubes

Each test requires two fluoride/oxalate (grey top) tubes.

PROCEDURE

The patient may drink additional water during the test and must be seated quietly throughout the test. Smoking is not permitted during the test.

The GTT procedure is described below. **Samples must be clearly labelled with patient name, date and time, e.g. 09:30**

TIME	BLOOD SAMPLE (Grey top sample for glucose)
Fasting sample	Minimum volume 1 mL (grey top)
0 hours	Polycal drink, followed by 100ml water, to be drunk within 5 minutes
2 hours sample	Minimum volume 1 mL (grey top)

INTERPRETATION

Normal

Both of the following criteria must be met:

- 1) Fasting glucose ≤ 6.0 mmol/L
- 2) 2 hour GTT glucose <7.8 mmol/L

Diagnostic Criteria for Diabetes Mellitus

Either of the following is diagnostic:

Classical symptoms of DM **and** any one of the following:

- a) fasting glucose ≥ 7.0 mmol/L
- b) random glucose ≥ 11.1 mmol/L
- c) 2 hour GTT glucose ≥ 11.1 mmol/L

Or, in an asymptomatic patient, any two of the following, obtained on separate occasions:

- a) fasting glucose ≥ 7.0 mmol/L
- b) random glucose ≥ 11.1 mmol/L
- c) 2 hour GTT glucose ≥ 11.1 mmol/L

Diagnostic Criteria for Impaired Glucose Tolerance

Both of the following criteria must be met:

- 1) Fasting glucose < 7.0 mmol/L
- 2) 2 hour GTT glucose ≥ 7.8 mmol/L but < 11.1 mmol/L

Diagnostic Criteria for Impaired Fasting Glycaemia

Both the following criteria must be met:

- 1) Fasting glucose > 6.0 but < 7.0 mmol/L
- 2) 2 hour GTT glucose < 7.8 mmol/L

Staging of type 1 diabetes

OGTT can be used to assist with staging of pre-symptomatic type 1 diabetes as follows:

- Normal OGTT - Stage 1
- Impaired Glucose Tolerance or Impaired Fasting Glycaemia - Stage 2
- Abnormal OGTT - Stage 3 or 4

Guidelines are currently developed around this and all results should be discussed with a consultant diabetologist

Equivocal Results

If the results of the GTT are equivocal the test should be repeated annually or sooner if the patient becomes symptomatic. A high index of suspicion should be maintained in a patient with equivocal results who also has a first degree relative with DM. Patients should not be falsely reassured that DM has been excluded when equivocal results are obtained.

References

1. American Diabetes Association – Report of the expert committee on the diagnosis and classification of diabetes mellitus. Diabetes Care 1997; 20: 1183-1197
2. WHO Expert Committee on Diabetes Mellitus. WHO Technical Report Series 2000

3. Recommendations for the Management of Diabetes. Diabetes UK October 2000.
www.diabetes.org.uk
4. Staging Presymptomatic Type 1 Diabetes: A Scientific Statement of JDRF, the Endocrine Society, and the American Diabetes Association. Diabetes Care 2015; 38(10):1964–1974
5. WHO Consultation report 2011. Use of Glycated Haemoglobin (HbA1c) in the diagnosis of Diabetes mellitus

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